

Saad Abdur Razzaq

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Summary

Machine Learning Engineer and AI researcher focused on developing intelligent, data-driven systems for real-world problems. Experienced in deep learning, NLP, computer vision, and applied AI automation using TensorFlow, PyTorch, and LangChain, with a track record of deploying production ML systems for real clients.

Education

National University of Computer and Emerging Sciences (NUCES-FAST) Sep 2020 – Jun 2024
B.Sc. Computer Science Chiniot-Faisalabad, Pakistan

- Relevant coursework: Data Structures, Artificial Intelligence, Numerical Computing, Software Engineering, Probability and Statistics, Applied Machine Learning, Fundamentals of Computer Vision

Pakistan International School Aug 2018 – Jun 2020
Higher Secondary School Certificate (HSSC) Al-Khobar, Saudi Arabia

Work Experience

Freelance Machine Learning Engineer Jun 2024 – Present
Engineered and deployed end-to-end AI/ML systems across education, healthcare, agri-tech, and intelligent automation

- Developed and deployed ML-driven automation systems for education platforms, focusing on intelligent grading, adaptive learning, and AI-assisted content creation
- Delivered fully functional grading and adaptive learning pipelines within 2 months for a client, accelerating product readiness and enabling early client onboarding
- Automated content generation and evaluation workflows within the first quarter, reducing manual workload for academic teams and streamlining deployment across multiple institutions
- Built an automated MCQ grading system using OCR/OMR pipelines for large-scale exam evaluation, cutting manual grading effort for an education-sector client
- Designed and deployed an edge-AI livestock monitoring platform for an agri-tech client, combining multi-camera vision, thermal sensing, and biosignal analytics on Jetson Orin hardware to detect health and behavioral patterns in real time
- Built an AI-powered ambient clinical scribe for a healthcare client, using automated speech recognition and language modeling to generate structured clinical documentation from patient-doctor conversations
- Delivered a bilingual PDF text-replacement and layout-preservation framework for a localization client, using OCR and NLP to cut manual localization time and cost while preserving document structure
- Developed a distributed classroom-intelligence system integrating multimodal audio, text, and engagement signals to give instructors real-time participation and comprehension insights
- *Tech stack:* Python, TensorFlow, PyTorch, LangChain, OpenAI GPT, spaCy, Pinecone, Whisper, BART, FastAPI, Docker, SQL, Pandas, NumPy, React, Node.js

Teaching Assistant – Data Science & Business Analytics Aug 2023 – Jun 2024
NUCES-FAST, CFD Campus Chiniot-Faisalabad, Pakistan

- Supported course delivery across three core subjects, designing assignments, quizzes, and lab exercises aligned with learning outcomes and assessment standards

- Assisted students in applying Python for data analysis, visualization, machine learning, and predictive modeling
- Guided learners in software refactoring, code quality improvement, and maintainability through code review sessions using Git and SonarQube

Technical Skills

- **Programming Languages:** Python, C, C++, C#, MATLAB, R
- **Databases:** MongoDB, MySQL, PostgreSQL
- **ML / DL Frameworks:** TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn
- **Data Science:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit
- **NLP:** Gensim, Flair, spaCy, TensorFlow Hub
- **Web / API:** FastAPI, React, Node.js, Express.js
- **Systems & DevOps:** GNU/Linux, SSH, PGP/GPG & PKI, Google Cloud, Docker, Jenkins, Gerrit, JFrog, Helm, Ansible

Portfolio Projects

- **AgroSentience** – Autonomous livestock intelligence platform integrating real-time vision, thermal sensing, and biosignal analytics to monitor cattle health and behavior, detecting estrus cycles, hunger, dehydration, lameness, and stress via multi-camera video streams and predictive models. (*Edge AI, Jetson Orin, YOLOv8, OpenCV, TensorRT, DeepSORT, MQTT, FastAPI, PyTorch, OpenVINO, LoRaWAN*)
- **SynapseOS** – On-device neural operating environment enabling persistent multi-agent collaboration, inter-process LLM communication, and real-time perception pipelines. (*PyTorch, Ray, gRPC, Vulkan, TensorRT*)
- **ClassMind** – Distributed classroom intelligence system integrating multimodal data (audio, text, sentiment, engagement) to autonomously evaluate participation and concept grasp, with real-time instructor insights. (*GPT-4, Whisper, BERT, LangGraph, TensorFlow, FastAPI, WebRTC*)
- **EffiScribe** – AI-powered ambient clinical scribe that records patient–doctor conversations and generates structured SOAP notes via automated speech recognition and language modeling. (*OpenAI Whisper, NLP, React Native, Next.js, FastAPI*)
- **AutoGrade** – Automated MCQ grading system for large-scale evaluation, reducing manual effort. (*Python, OpenCV, OCR/OMR, FastAPI*)
- **Notive AI** – Cross-platform, privacy-first voice note-taking app with on-device ASR for multilingual transcription. (*Whisper AI, TensorFlow, LiveKit, React Native*)

Certifications

- IBM – *AI Developer* Professional Certificate (10 courses), 2025
- Google – *Advanced Data Analytics* Professional Certificate (8 courses), 2025
- IBM – *Data Science* Professional Certificate (12 courses), 2025
- IBM – *Generative AI Engineering* Professional Certificate (16 courses), 2025
- IBM – *Data Engineering* Professional Certificate (16 courses), 2025
- IBM – *AI Product Manager* Professional Certificate, 2025
- University of Pennsylvania – *AI For Business* Specialization, 2025
- ZTM Academy – *TensorFlow for Deep Learning* Bootcamp (63 hrs), 2025

Languages

Urdu – Native | English – C1 (IELTS Academic: 7.0 overall, no band below 6.5)